

WATERLOO, IA, USA

WMO: 725480

Lat: 42.554N

Lon: 92.401W

Elev: 868

StdP: 14.24

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94910

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.9	(c) -4.7	(d) -16.5	(e) 2.3	(f) -8.8	(g) -11.6	(h) 3.0	(i) -3.9	(j) 28.8	(k) 19.1	(l) 26.4	(m) 17.8	(n) 6.9	(o) 300	(p) 0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	90.9	75.5	88.1	73.8	85.5	72.3	78.6	86.9	76.6	84.5	74.8	82.5	12.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.2	141.1	84.5	74.1	131.2	81.9	72.1	122.8	79.9	42.7	86.8	40.6	84.6	38.8	82.6	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 25.9	(b) 23.2	(c) 20.1	(e) -19.5	(f) 95.5	(g) 7.5	(h) 3.0	(i) -24.9	(j) 97.6	(k) -29.3	(l) 99.4	(m) -33.5	(n) 101.1	(o) -38.9	(p) 103.3		
(4) 25.9	(5) 23.2	(6) 20.1	(7) -19.8	(8) 81.4	(9) 7.4	(10) 2.1	(11) -25.1	(12) 82.9	(13) -29.4	(14) 84.1	(15) -33.6	(16) 85.3	(17) -39.0	(18) 86.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	18.2	22.6	35.4	48.4	60.2	70.2	73.2	70.8	63.9	50.7	36.7	24.2
	DBStd	21.59	13.01	12.83	12.16	9.78	8.42	6.27	5.71	5.43	8.18	9.55	10.58	12.06
	HDD50	3708	984	768	476	140	13	0	0	0	4	108	415	800
	HDD65	6977	1449	1187	919	502	198	23	5	11	117	453	849	1264
	CDD50	2986	0	1	23	93	329	606	720	644	422	131	16	1
	CDD65	781	0	0	1	6	49	179	260	191	85	10	0	0
	CDH74	7459	0	0	12	101	518	1668	2517	1665	861	115	2	0
	CDH80	2404	0	0	0	24	144	561	892	489	268	26	0	0
	WSAvg	9.5	10.3	10.6	10.9	11.8	10.3	8.6	7.4	6.8	8.0	9.4	10.1	9.8
	PrecAvg	34.4	0.7	1.1	2.3	3.5	4.6	4.4	4.9	3.9	3.2	2.6	1.9	1.2
(15) Precipitation	PrecMax	53.1	1.8	3.5	5.4	8.5	11.3	10.1	12.8	10.6	11.4	5.4	5.6	3.8
	PrecMin	19.0	0.1	0.0	0.2	0.9	0.8	1.4	0.3	0.1	0.6	0.0	0.1	0.1
	PrecStd	7.5	0.5	0.8	1.2	1.7	2.4	2.0	3.1	2.5	2.4	1.4	1.4	0.8
	PrecStd	7.5	0.5	0.8	1.2	1.7	2.4	2.0	3.1	2.5	2.4	1.4	1.4	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.2	57.0	75.0	83.7	89.4	92.8	95.9	92.5	90.7	83.6	70.4	57.7
		MCWB	42.9	49.4	60.7	64.0	71.3	75.7	78.0	75.2	73.1	67.6	56.9	52.4
	2%	DB	42.7	49.6	66.3	76.2	83.9	89.1	91.2	88.5	86.5	76.7	63.2	49.5
		MCWB	38.2	44.2	55.5	60.0	66.8	73.1	77.6	75.3	71.0	62.7	54.7	44.1
	5%	DB	38.2	43.5	59.9	70.7	79.9	86.3	88.0	85.1	83.0	71.7	57.5	44.5
		MCWB	35.2	39.4	51.9	56.4	64.6	71.4	75.7	73.4	68.8	61.3	50.1	40.6
	10%	DB	34.7	38.8	53.0	65.9	75.7	83.0	85.0	82.3	79.1	66.9	52.7	39.4
		MCWB	32.6	35.7	46.4	53.8	62.5	69.5	73.9	71.8	66.6	58.1	46.3	36.4
	0.4%	WB	43.7	50.5	62.6	65.9	73.8	79.5	82.0	80.1	76.1	70.3	60.2	54.1
		MCDB	50.4	56.2	71.8	78.6	84.9	89.8	90.7	88.0	86.5	78.2	67.2	56.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.9	44.6	58.3	62.1	70.1	76.2	79.5	77.4	73.4	66.4	55.9	45.9
		MCDB	42.5	49.0	64.4	72.6	80.1	85.0	87.9	85.3	82.7	72.9	61.5	48.4
	5%	WB	35.3	39.3	52.0	58.9	67.5	74.0	77.5	75.4	71.3	62.8	51.1	40.3
		MCDB	37.8	43.0	58.9	68.6	76.4	83.0	85.2	82.3	78.9	69.5	56.9	44.0
	10%	WB	32.9	35.9	46.3	54.9	64.8	71.8	75.4	73.5	69.0	59.2	46.8	36.4
		MCDB	34.6	38.4	52.4	63.7	72.8	80.6	82.9	79.9	76.2	65.8	52.0	39.3
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	16.5	18.4	21.7	21.1	20.0	19.9	20.3	23.3	21.3	18.4	15.6
		MCDBR	20.4	20.4	26.9	30.3	26.1	23.4	21.7	22.2	26.4	27.0	26.2	20.9
		MCWBR	16.4	15.5	16.7	16.8	12.6	10.8	10.6	11.2	12.5	15.1	17.5	16.4
	5% WB	MCDBR	18.8	19.0	24.8	25.7	22.2	20.3	19.7	19.1	21.3	22.3	23.3	19.3
		MCWBR	15.4	14.9	16.2	16.1	12.2	11.2	10.8	10.9	11.5	14.9	17.0	15.9
(40) Clear-Sky Solar Irradiance	taub		0.284	0.309	0.332	0.376	0.406	0.418	0.434	0.415	0.384	0.340	0.301	0.279
	taud		2.379	2.313	2.372	2.281	2.266	2.276	2.215	2.281	2.328	2.446	2.481	2.421
	Ebn at Noon		273	282	289	282	276	272	266	267	268	268	266	265
	Edh at Noon		25	32	34	40	42	42	44	40	36	28	23	22
	RadAvg		586	916	1220	1482	1762	1944	1977	1714	1411	923	624	463
(45) All-Sky Solar Radiation	RadStd		57	75	128	132	122	167	115	110	102	118	51	34

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page